

Turning Uncertainty Into Knowledge

Inverse Problem Theory Lifted to the Computational Top-Level

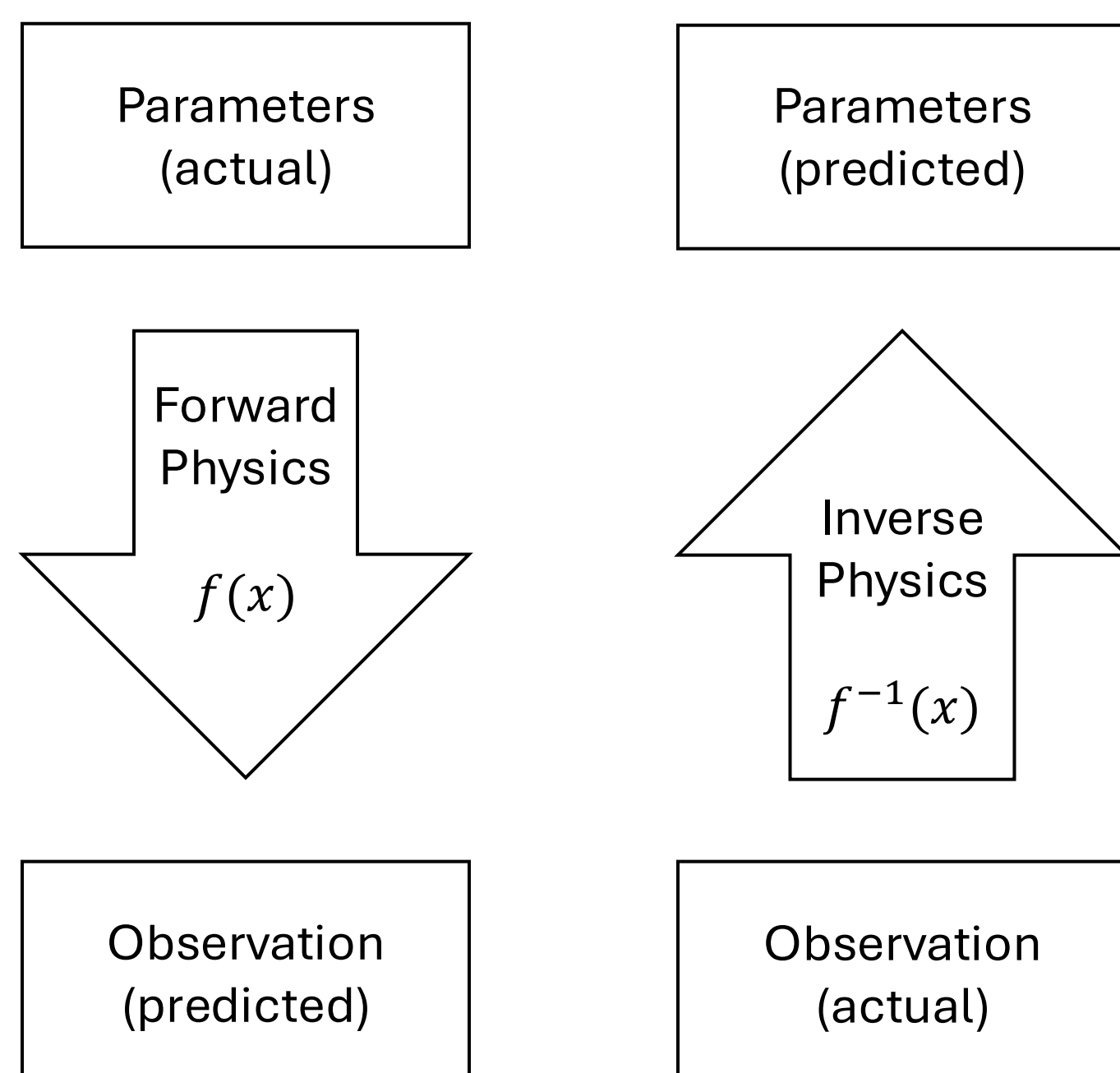
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Inverse problems in remote sensing are computationally expensive and typically lack exact uncertainty tracking. **We apply machine learning frameworks for automatic differentiation of forward model physics**, promising accelerated parameter retrieval and mathematically rigorous uncertainty propagation at tensor-level.

The Remote Sensing Blind Spot

1. The Inversion Dilemma

We can easily predict observations from given physical parameters. But our real job is the reverse: predicting physical parameters from given observations. Going backward is our blind spot (and pain point) because the forwards physics, like, e. g., radiative transfer, is fundamentally not invertible.

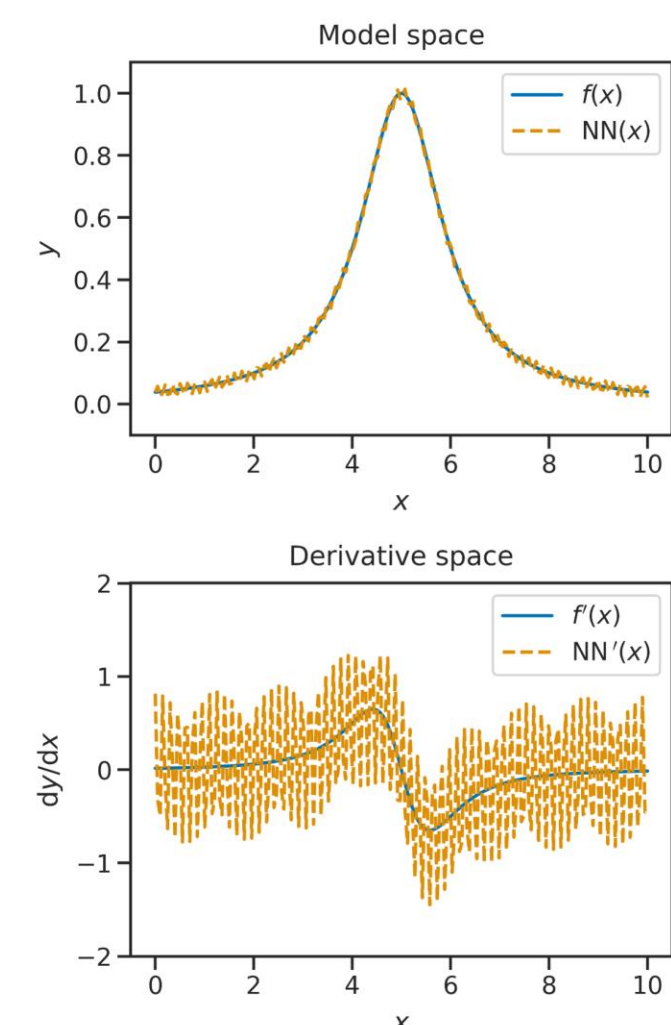


2. Conventional Approaches and Why They Fail

- **Finite difference approximations (pure physics):** Way too slow: $O(n)$ forward runs per iteration. Highly unstable convergence due to only approximate derivatives. Inaccurate uncertainty propagation.
- **Forward model emulators (pure ML):** Fast approximation of forward physics but poor approximation of derivatives; highly unstable convergence. Uncertainty propagation impossible.
- **Direct neural inversion (pure ML):** Meaningless if the inverse problem is inherently ill-posed. Yields physically inconsistent results. Uncertainty propagation is impossible.

Fun fact: Purely data-driven

neural networks approximate target functions with minimal residual error. Nevertheless, their derivatives with respect to inputs can completely blow up, as standard loss functions do not regularize the gradient space. **Sobolev training** fixes this by learning both the target function and its gradients.



Differentiable Programming (∂P)

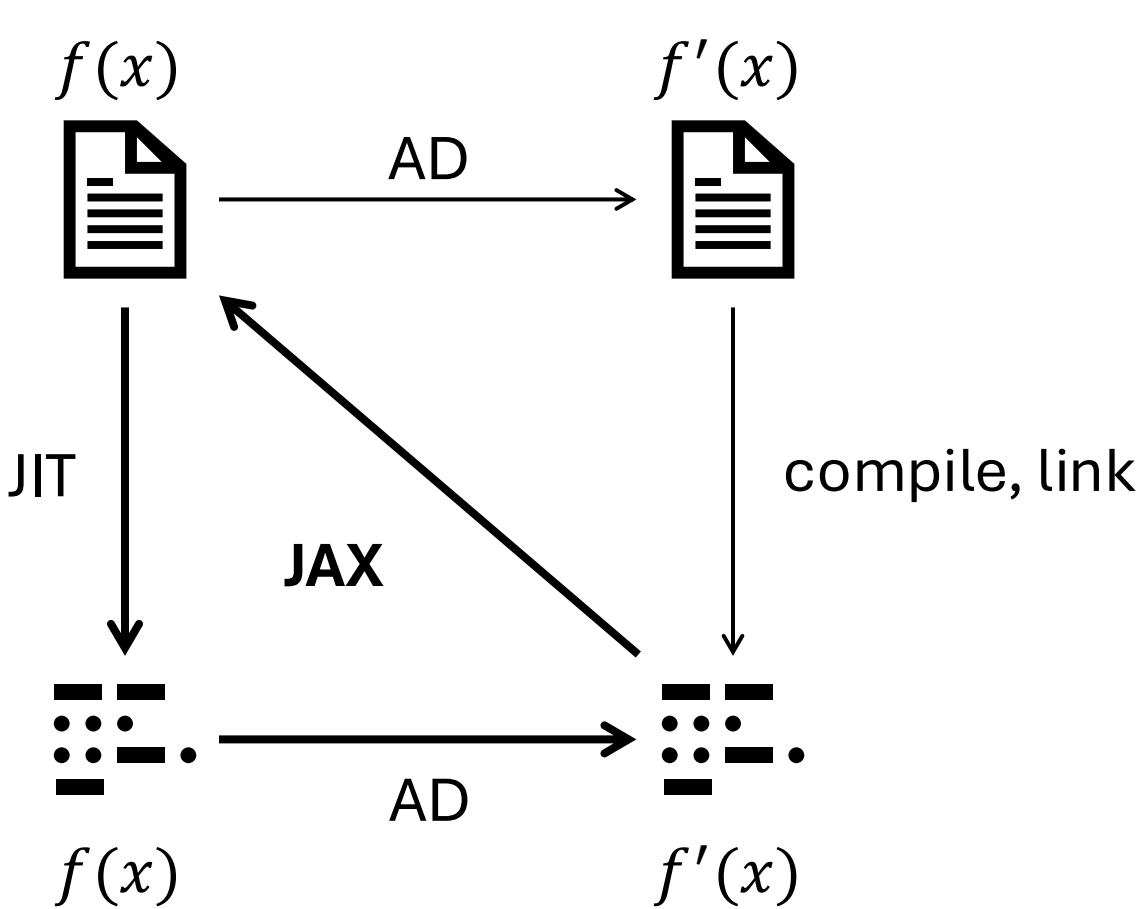
3. The Computing Evolution

- **The concept: We inject physics into AI infrastructure.** Unlike purely data-driven ML, we do **not** ignore our knowledge on physics for AI.
- **The mechanism:** Parameters in physical forward models are optimized exactly like neural network weights.
- **The power of JAX:** Utilizing Google's JIT compiler and automatic differentiation, model-to-observation misfits are backpropagated directly through the physics code.
- **The breakthrough:** The availability of exact gradients greatly facilitates the solution of the inverse problem with $O(1)$ complexity for n parameters per iteration—fast, stable, and physically consistent.
- **The bonus:** The mathematical law of propagation of uncertainty is built-in directly at the tensor level, executing seamlessly as part of the core computational graph.

4. The Programming Workflow

User writes pure code: The workflow begins with the programmer writing standard, stateless Python code using NumPy-like syntax to express the forward physics equations. The physics are implemented purely and explicitly.

Declaring derivatives: Within his code, the user can directly request derivatives of any order (e.g., Jacobians or higher-order derivatives) as regular programmatic expressions.

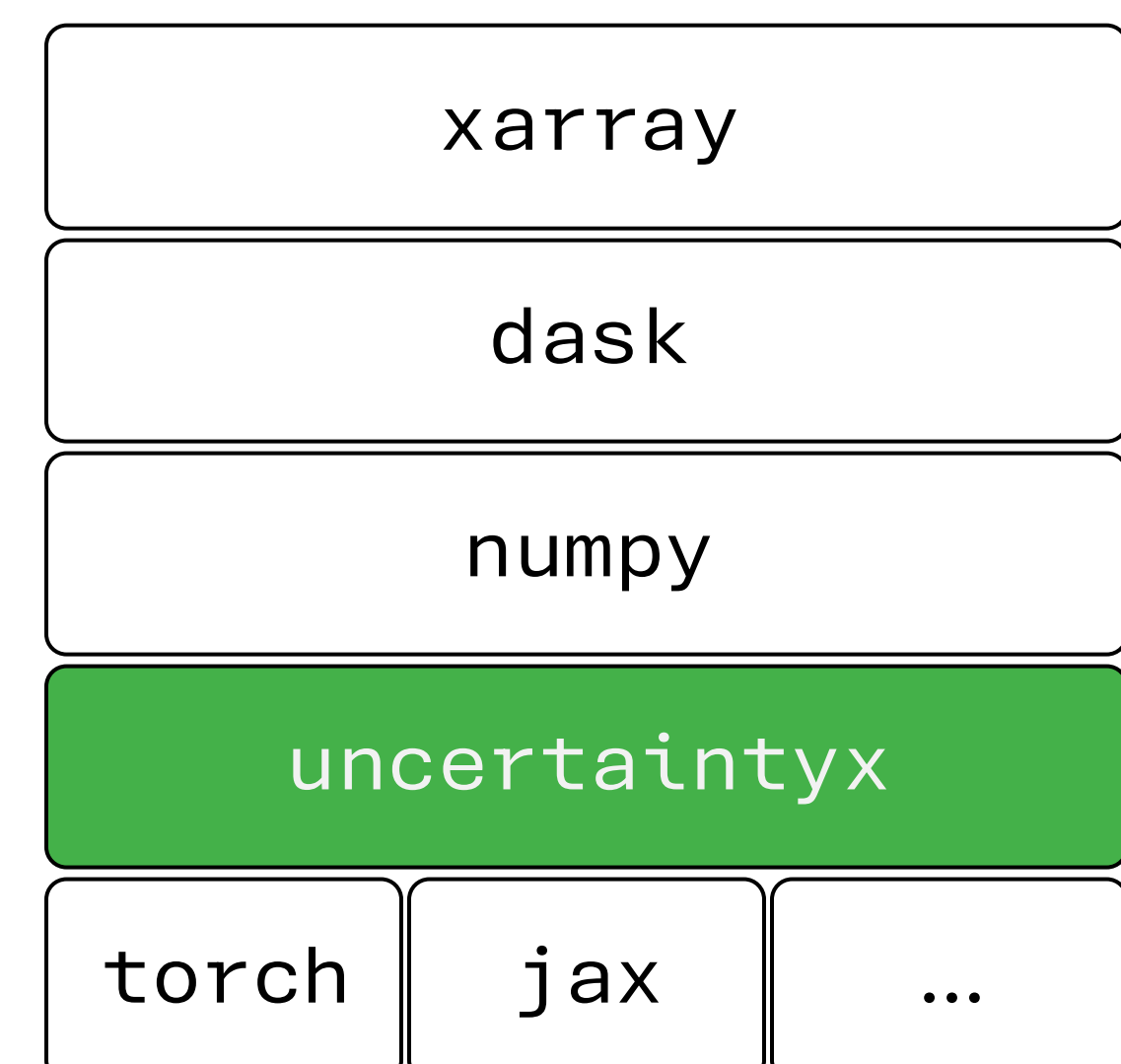


Just-in-time (JIT) compilation: Unlike source-to-source automatic differentiation (AD), which statically transforms code text, JAX traces the execution path at runtime. When the code is executed, the JIT compiler instantly processes the entire program and generates highly optimized machine code for both the forward physical equations and any requested analytical derivatives simultaneously. JAX is a framework for high-performance array computing across CPUs, GPUs, and TPUs, turning derivatives from external annotations into active elements of the data-processing code.

Code and Applications

5. Uncertaintyx

Uncertaintyx (or just Tyx) is a lightweight, open-source Python layer developed by Brockmann Consult that provides a simple façade for JAX and PyTorch to encapsulate the complexities of tensor-level differentiable programming. Tyx provides custom @jitted functions for batch vectorization, Jacobians, Hessians, optimal estimation (OE) and errors-in-variables (EIV) inversion algorithms and embeds the mathematical law of propagation of uncertainty directly at the tensor level. Tyx facilitates seamless interoperability with native numerical Python libraries like NumPy, Dask, and Xarray.



6. A Solution in Search of a Problem

The development of Uncertaintyx (Tyx) was directly driven by requirements within ESA's Climate Change Initiative (CCI). Current and possible future target applications address distinct computational and metrological tasks across major Earth Observation frameworks:

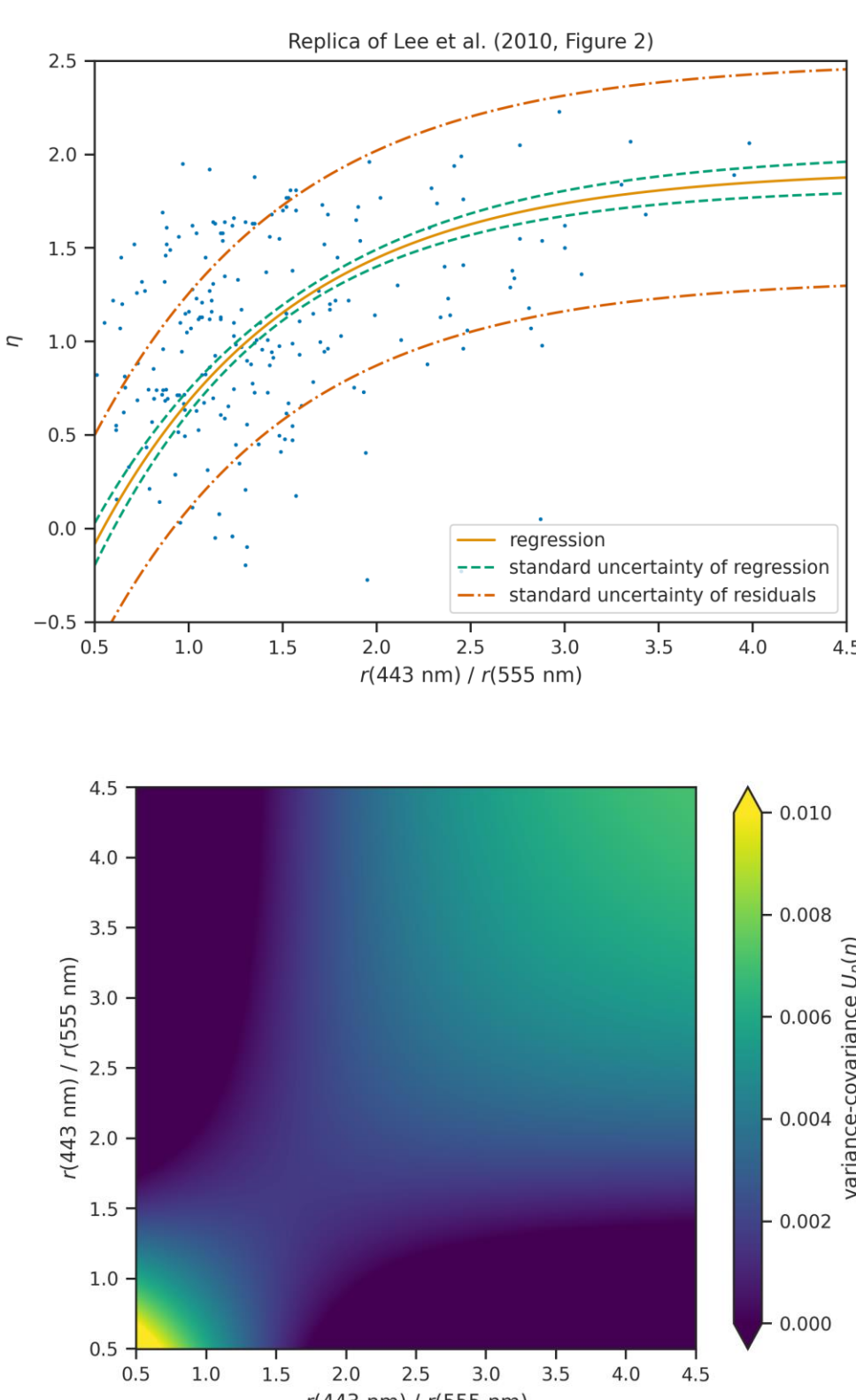
- **Ocean Colour & Phytoplankton CCI:** Computes the mathematically rigorous uncertainty budgets required for Essential Climate Variables (ECVs).
- **Land Cover CCI:** Inversion and uncertainty budgeting of Sentinel-3 atmospheric correction and seasonal surface reflectance products.
- **Copernicus CO2M & EUMETSAT Sentinel-3 Synergy:** Inversion of radiative transfer equations to retrieve cloud-effective obstruction products.
- **Mission Performance Centres:** A candidate to revive the multi-mission AD-based calibration workflows from the FIDUCEO project to underpin global climate data records (Giering et al., 2019; Quast et al., 2019).

Giering et al. (2019) A Novel Framework to Harmonise Satellite Data Series for Climate Applications, doi.org/10.3390/rs11091002. **Quast et al. (2019)** Climate Data Records from Meteosat First Generation Part II: Retrieval of the In-Flight Visible Spectral Response, doi.org/10.3390/rs11050480.

Analytical benchmarks within the Uncertaintyx

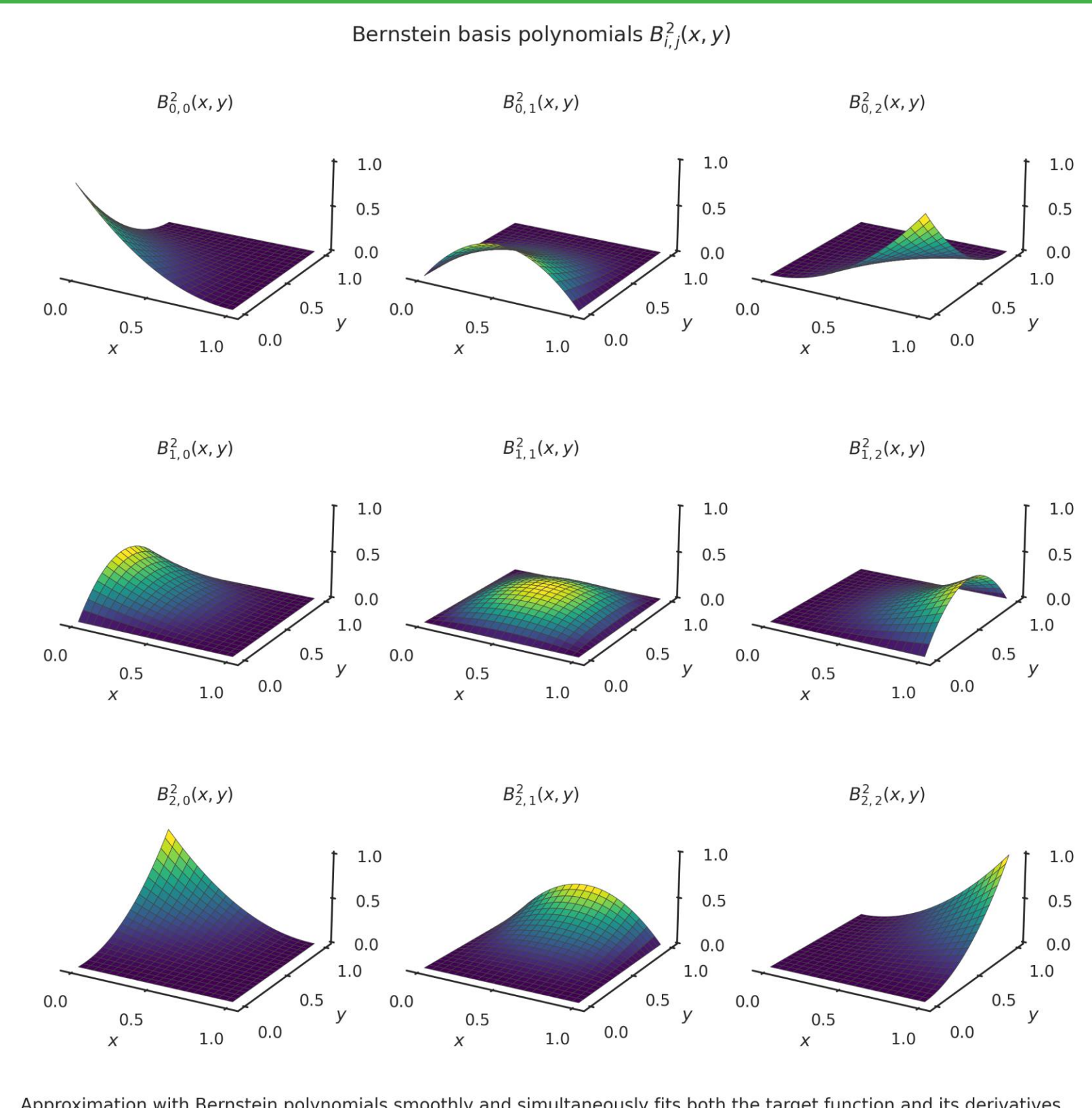
unit-level test suite: Replica of the empirical regression from Lee et al. (2010, Figure 2) showing the parameter η against the blue-to-green reflectance ratio, along with the standard uncertainty of the regression (dashed) and residuals (dash-dotted). The variance-covariance matrix (lower panel) analytically propagated by Uncertaintyx, serves as a baseline component for the full in-water inverse problem.

Lee et al. (2002) Deriving inherent optical properties from water color: a multiband quasi-analytical algorithm for optically deep waters, doi.org/10.1364/AO.41.005755. **Lee et al. (2010)** An Update of the Quasi-Analytical Algorithm.



Universal differentiable N-D

lookup tables: Bernstein basis polynomials provide an implicit **Sobolev approximation** to transform arbitrary multivariate lookup tables into continuously differentiable forward operators, guaranteeing exact gradients for inversion.



Development Status: Alpha
Programming Language: Python
Topic: Scientific/Engineering – Information Analysis
Code Repository: github.com/bcdev/uncertaintyx
License: MIT